

MSC 488

Interactive Computer Music with Max

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Class Hours: Tuesday 8:30-10:20, Thursday 13:30-15:20 (room: 336/341)

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COURSE DESCRIPTION: Hands-on programming and composition course that explores sound design and HCI (human-computer interaction) as the ingredients for creating engaging real-time interactive musical experiences. Students will learn about the principles, design considerations, and aesthetic qualities of interactive music through the Max environment.

Although the course will touch upon theory, the focus is on sonic and compositional aspects. We will work primarily with the Cycling 74 Max software. A good amount of work will be spent on developing a rigorous understanding of Max and creating a work of art using this technology. We will cover a variety of techniques such as filtering, pitch shifting (including harmonization, FFT analysis/resynthesis, and granulation), a variety of synthesis techniques, chorusing, looping, nonlinear wave shaping, distortion, reverb, delay, spatialization, or a myriad other possible digital effects.

What this course is not about:

- Making beats, grooves and loops
- Learning to mix your songs
- Learning about music production and engineering
- Learning to make music with software instruments
- Learning music theory to be able to make better songs
- Producing songs
- Songwriting

What this course is about:

- Composing and realizing creative music pieces that are based on human-computer interaction.
- Developing critical listening skills and broadening your idea of music.
- Building an in-depth knowledge of live interactive music
- Learning fundamentals of acoustic and electroacoustic music theory in order to compose interactive music.
- Learning the nuts and bolts of the Max environment and being able to develop complex Max patches.
- Learning to work with audio and midi processing algorithms

COURSE STRUCTURE.

This is a high-intensity course structured around three modules.

1. Basic Theory: Although this is primarily a practice-oriented course, a deeper understanding of the concepts is indispensable when working with the electroacoustic medium.
2. Practice: Our primary purpose of the class is to produce interactive music through Max. Students are expected to devote a considerable amount of patching every week. The first two-thirds of the semester will be spent on the nuts and bolts of Max software. We will focus on individual projects for the remaining third of the semester.
3. Besides compositional activity, there will also be an emphasis on the aesthetics and culture of electroacoustic music. We will have a weekly listening list and spend about half an hour each week discussing various aspects of the genre.

REQUIRED TEXTBOOK

Cipriani & Giri, *Electronic Music, and Sound Design: Theory and Practice with Max 8 Volumes 1-3*

You must get a copy of the textbook. We will strictly follow these two volumes for the first two-thirds of the semester. I will post other readings on moodle side. We will also have a google drive folder for all the patches etc.

https://drive.google.com/drive/folders/1753_qGd8-OPof6cjcQQGYdszYozyEre

Please also download all the supporting files for the textbooks from:

<http://virtual-sound.com/cmsupport>

<https://virtual-sound.com/emasd-2-support/>

REQUIRED SOFTWARE

You must have Max installed on your computer. Max 9 is the latest version but, Max 7 or 8 would also do it.

Max is installed on the studio computer; however, you will need it on your personal computer. If you don't plan on purchasing the software, I suggest not taking the course. There's a special student offer for 108\$/year. Or a regular subscription for 15.99\$/month. For detailed information, please visit:

<https://cycling74.com/shop>

<https://cycling74.com/shop/academic/max>

REQUIRED MATERIALS

In each class, you need to bring the following:

- A computer with Max 9 installed
- A pair of headphones

OTHER RECOMMENDED TEXTS.

Roads, Curtis. *The computer music tutorial*. MIT Press, 1996.

Loy, Gareth. *Musimathics: the mathematical foundations of music*. Vol. 1. MIT Press, 2011

Holmes, Thomas B., and Thom Holmes. *Electronic and experimental music: pioneers in technology and composition*. Psychology Press, 2002.

RECOMMENDED TUTORIALS

<https://www.kadenze.com/courses/programming-max-structuring-interactive-software-for-digital-arts/info>

GRADING.

Presentation: %5

Max assignments (6): %30 (%5 each)

Midterm Max Assignment: %17.5

Take-home Midterm Theoretical Exam: %12.5

Final Project %25

Attendance and Participation: %10

WEEKLY SCHEDULE:

		COURSE	SUBJECT	ASSIGNMENT S DUE	READINGS
1	16-9	Introduction to Human-Computer Interaction and Max Environment Getting started, first patches	Opening Max, patchers, Audio-Midi connections, making the first sound		
2	18-9	Chapter 1 cont.	Making first VCO and env.		Chapter 1T and 1P
3	23-9	Max Essentials (Int. A)	Int. A	Assignment #1	Chapter: Interlude A
4	25-9	Additive and Vector Synthesis (2P)	Additive, wavetable synth, phase, spectrum, beats		Chapter 2T and 2P
5	30-9	Additive and Vector Synthesis cont.	Harmonicity, OSC bank,	Assignment #2	
6	2-10	Noise Generators, Filters and Subtractive synthesis (3P)	Noise, various filter types, Q, parallel vs serial, filter orders		Chapter 3T and 3P
7	7-10	Noise Gen., Filters and Subtractive synthesis (3P)	Subtractive synth, FIR vs IIR, Graphic EQ, Resonators	Assignment #3	
8	9-10	Additional Elements (Int. B)	Int. B		Chapter: Interlude B
9	14-10	Control Signals (4P)	LFO, DC offset, amplitude-frequency modulations, Modulation synthesis.		Chapter 4T and 4P
10	16-10	Control signals cont. and connecting MIDI		Assignment #4	
11	21-10	Digital Audio Working with sampled sounds (5P)			Chapter 5T and 5P
12	23-10	Polybuffer~ / Building a simple sampler			Chapter: Interlude C

13	30-10	Midterm Exam!			
14	4-11	Class Presentation of Monophonic Synthesizers		Midterm Project	
15	6-11	Polyphony (Int. C)			Chapter 6T and 6P
16	11-11	Building a polyphonic sampler		Assignment #5	Chapter 7T and 7P
17	13-11	Delay (6P)			Chapter: Interlude D
18	18-11	Dynamic processors (7P)			TBA
19	20-11	Pitch and attack detection		Assignment #6	
20	25-11	Non-linear Sythesis			
21	27-11	Preset and Patcher management (Int D)			
22	2-12	FFT			
23	4-12	FFT cont.			
24	9-12	Multichannel audio			
25	11-12	Building a Granular Synthesizer			
26	16-12	Putting everything together / Building a concert patch			
27	18-12	Projects			
28	23-12	Projects			
	25-12			Final Project (score and patch)	

Weekly preparation and assignments. Every class you will be assigned readings from Cipriani and Giri but there will also be reading assignments from articles and software documentation as well as a small project consisting of a Max software patch demonstrating understanding of the week's material. In certain weeks, you will be asked to compose music with certain constraints.

All assignments must be uploaded to the drive folder.

Midterm Max Assignment

Build a simple polyphonic synthesizer with Max. You can use any type of controller.

Presentation: You need to present a piece of music during class. There will be a repertoire. You can pick the piece. You are expected to do a detailed presentation.

Attendance/Class participation

Class participation is mandatory. This is not a general core class but a specialty class so you are expected to come to every meeting with the exception of emergencies of

course. I can tolerate up to 6-hours of absence without any grade drop. If your absences pass beyond this limit, your attendance grade will be affected, according to the table below. Please try to be prompt as well. Late arrivals (after the first 15 minutes) will count as a 1-hour absence. After a total of 18-hour absence, you will receive a letter grade of FZ.

I also expect you to come to class prepared, with the material in hand. This will directly affect your participation grade.

This is a fun class, show up, learn stuff, and you'll do just fine!

1 absence (2 hours)	15 points
2 absences (4 hours)	15 points
3 absences (6 hours)	15 points
4 absences (8 hours)	14 points
5 absences (10 hours)	12 points
6 absences (12 hours)	9 points
7 absences (14 hours)	5 points
8 absences (16 hours)	0 points
9 absences (18 hours)	FZ

Late work policy

Assignments are due as indicated in class. Assignments turned in after the due date will be received %50 of the grade. If they exceed three days, an automatic F will be given.

Final Project

For the final, each student will create a composed interactive piece using a non-standard instrument (e.g. toy, found object, or unconventional acoustic source) and a self-built Max patch. The system must respond to live audio input only—no MIDI or external Max objects are allowed. Students are required to submit a written score (in any interpretable format) alongside a 5-7 minute performance of the work. The final piece must be performable in a live concert setting and demonstrate meaningful, real-time interaction between performer and software.

Academic Integrity

Bilkent University is committed to fundamental values necessary to research, academic excellence, the pursuit of learning, and a culture of integrity. These include honesty, trust, fairness, respect, and responsibility among all members of the University community. Faculty, instructors, staff, and students are all strongly encouraged to embrace these values in pursuit of these shared goals.

Academic Dishonesty. In light of its commitment to academic integrity, Bilkent University prohibits acts of misconduct and academic dishonesty. These include, but are not limited to, acts of cheating, plagiarism, and falsification of data, as defined below. Cheating occurs when an individual uses dishonesty or deception to receive or help others receive professional or academic credit for work she or he did not perform. Cheating includes, among other acts, misappropriation and / or development of the ideas, concepts, designs, or methodology of others without consent; use of materials or devices not permitted by the instructor during exams; taking an exam for another person; resubmitting work previously submitted elsewhere; copying previously published solutions to problems.

Plagiarism is representing the work or ideas of another person as one's own. It frequently involves quoting, cutting / pasting or closely paraphrasing written language without appropriately citing the source of the material through the use of quotation marks, reference notes, or other methods of acknowledgement. An act of plagiarism may be unintentional, and to avoid unintentional plagiarism standard practices of citation should be followed. For detailed instructions regarding standard citation practices, see <http://www.plagiarism.org>.

Falsification is a deliberate misrepresentation in which information, whether in the form of data, written language, images or other media, is either altered or fabricated.

Honesty and the University Honor Code which can be found at the following web address: <https://w3.bilkent.edu.tr/bilkent/policy-on-conflicts-of-interest-and-commitment-academic-integrity/>